SCHEME OF VALUATION - QP CODE: 25900198

MGU-UGP (HONOURS) REGULAR EXAMINATION-MARCH 2025

SECOND SEMESTER

MG2DSCSTA100-INTRODUCTION TO STATISTICAL MODELLING

PART A

Answer any **7** questions. Each question carries **2** marks

1. $\sum f(x) = 1$, K = 1/36 ---- 2 marks (The question contains a typographical error, so 2 marks may be given for the first step itself).
2. Two properties ----2 marks
   (Definition of mgf / any one property ---- 1 mark)
3. np = 5, npq =0.3 ----1 mark (or mean =np ,var = npq --- give 1 mark)
   pmf of binomial distribution----½ mark
   p = 0.94 ---½ mark
4. x=y/3 ----1 mark ,

   $f(x) = 6\left(\frac{y}{3}\right)\left(1-\frac{y}{3}\right)$ or $f(y) = f(x)\left|\frac{dx}{dy}\right|$ or $f(y) = \frac{2y(3-y)}{9}$, 0 < y < 3 ----1 mark

5. Formula of expectation ----1.5 marks
   E(X)= 2.23 ≈ 2 ----½ mark
6. p = 0.6, P(he misses the throw) = 1-p = 0.4 ----2 marks
7. Discrete r.v ----2 marks
8. P.d.f of Normal distribution /standard normal distribution ----2 marks
9. The last digit X can be any of the 10 digits (0 to 9) or X = 0,1,2,3,…,9 ----1 mark
   $P(X = 6) = \frac{1}{10}$ ----1 mark
10. Writing the pdf of Normal distribution with mean=100 and var = 25 ---- 2 marks
    (p.d.f of Normal distribution /standard normal distribution only ----1 mark)

PART B

Answer any **4** questions. Each question carries **6** marks

11. (1) $\int f(x)dx = 1$ ---- 2 marks

    $\int_0^1 kx^2\,dx = 1$ ---- ½ mark, k = 3 ---- ½ mark

    (2) $\int f(x)dx = 1$ ---- 2 marks

    $\int_0^1 kx(1-x)\,dx = 1$ ----½ mark, k = 6 ---- ½ mark

12. Formula of E(X) and V(X)----1 mark each

E(X) =$\frac{2}{3}$ ---- 2 marks , E(X^2) =$\frac{1}{2}$ ---- 1 mark

V(X) =$\frac{1}{18}$ ---- 1 mark

13. np = 6 ----1 mark, npq = 3.6 ----1 mark

q = 0.6 or p = 0.4 ---- 1 mark

$n = \frac{6}{0.4} = 15$ ---- 1 mark

$X \sim B(15,0.4)$ or writing the pmf of binomial distribution ---- 1 mark

$P(X = 2) = \binom{15}{2}(0.4)^2(0.6)^{13}$ or 0.0219 ---- 1 mark

14. Uses /examples or explanation of situations in which Poisson distribution is used or definition /mean and variance ----6 marks
15. Definition of exponential distribution ----5 marks, properties or giving mean and variance---- 1 mark
16. Definition of geometric distribution ----5 marks, properties or giving mean and variance---- 1 mark

PART C

Answer any **1** question. Each question carries **12** marks

17. Let X be the r.v denoting the number shown by an unbiased die.

p.m.f. of X, $P(X = x) = \frac{1}{6}$, $x = 1,2,3,4,5,6$ ---- 6 marks

distribution function definition ---- 3 marks

Computation of d.f.---- 3 marks

18. p = $\frac{1}{500}$, n= 40, N = 20000 ----2 marks

p.m.f. of Poisson distribution ---- 3 marks

$\lambda = np = \frac{4}{50} = 0.08$ ---- 2 marks

$p(x) = P(X = x) = \frac{e^{-0.08}(0.08)^x}{x!}$ ---- 2 marks

In a consignment of 20000 packets,number of packets containing x defectives is $N\,p(x)$.

(1) $N\,P(X = 0) = 20000 \times 0.9231 = 18462$ ---- 1 mark

(2) $N\,P(X = 1) = 20000 \times 0.0738 = 1476$ ---- 1mark

(3) $N\,P(X = 3) = 20000 \times 0.0000788 = 1.58 \approx 2$ ---- 1 mark